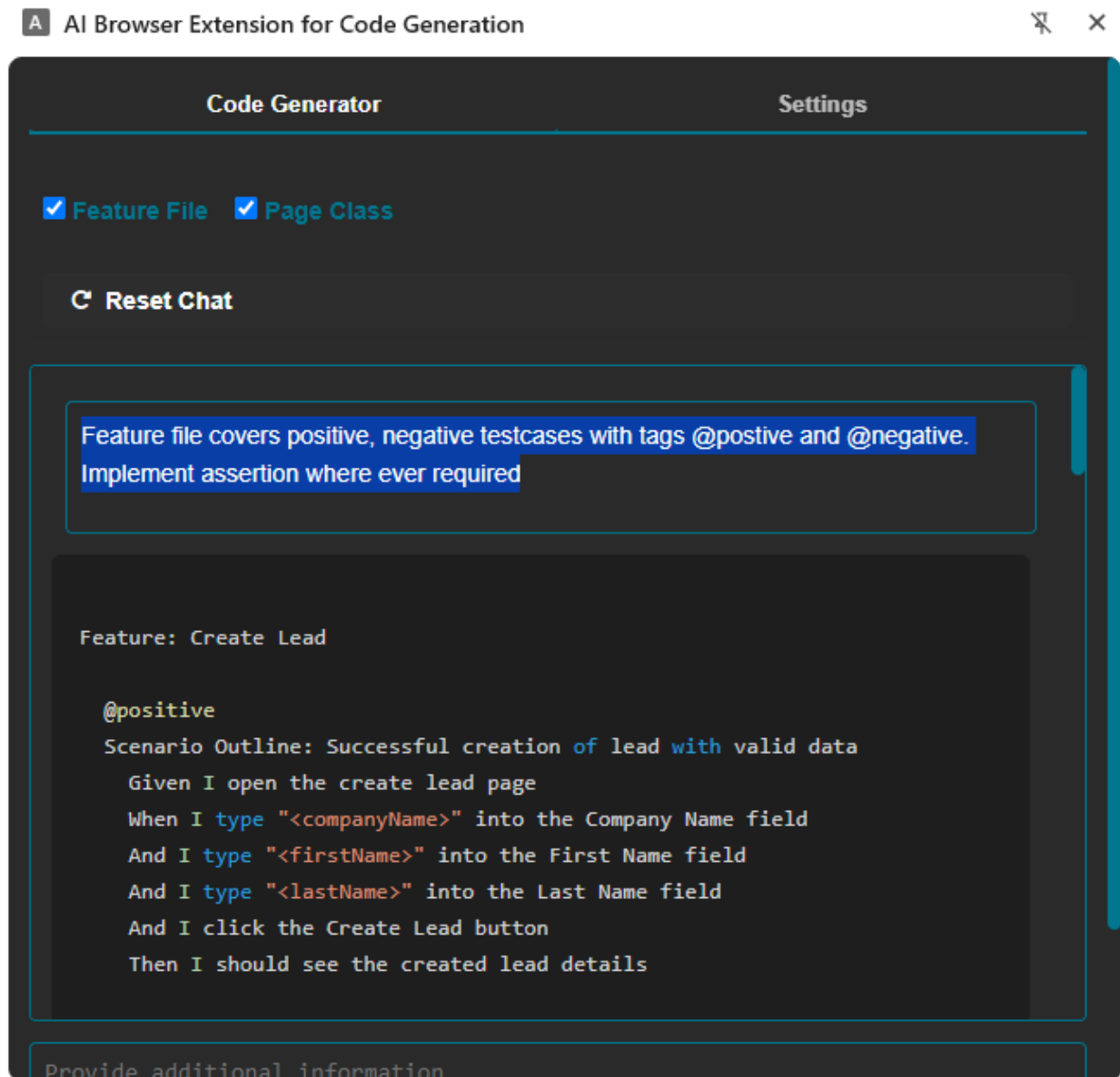


AI extension prompt & Testcases Generator

Feature file covers positive, negative testcases with tags @positive and @negative. Implement assertion where ever required



Feature: Create Lead

@positive

Scenario Outline: Successful creation of lead with valid data

Given I open the create lead page

When I type "<companyName>" into the Company Name field

And I type "<firstName>" into the First Name field

And I type "<lastName>" into the Last Name field

And I click the Create Lead button

Then I should see the created lead details

Examples:

companyName	firstName	lastName
Infosys	Ramesh	Kumar
Wipro	Suresh	Rao
Tata Consultancy	Maresh	Reddy

@negative

Scenario Outline: Unsuccessful creation of lead with invalid data

Given I open the create lead page

When I type "<companyName>" into the Company Name field

And I type "<firstName>" into the First Name field

And I type "<lastName>" into the Last Name field

And I click the Create Lead button

Then I should see an error message

Examples:

companyName	firstName	lastName
	Ramesh	Kumar
Wipro		Rao

```
package com.leaf taps.stepdefs;
```

```
import io.cucumber.java.en.*;
import org.openqa.selenium.*;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.support.ui.*;
```

```
import static org.junit.Assert.assertTrue;
```

```
public class CreateLeadStepDefinitions {
    private WebDriver driver;
    private WebDriverWait wait;
```

```
@io.cucumber.java.Before
```

```
public void setUp() {
    driver = new ChromeDriver();
    wait = new WebDriverWait(driver, java.time.Duration.ofSeconds(10));
    driver.manage().window().maximize();
}
```

```
@io.cucumber.java.After
```

```
public void tearDown() {
    if (driver != null) driver.quit();
}
```

```
@Given("I open the create lead page")
```

```
public void openCreateLeadPage() {
    driver.get("https://leaf taps.com/crmsfa/control/createLeadForm");
}
```

```
@When("I type {string} into the Company Name field")
public void enterCompanyName(String companyName) {
    WebElement el =
wait.until(ExpectedConditions.elementToBeClickable(By.id("createLeadForm_companyName")));
    el.sendKeys(companyName);
}

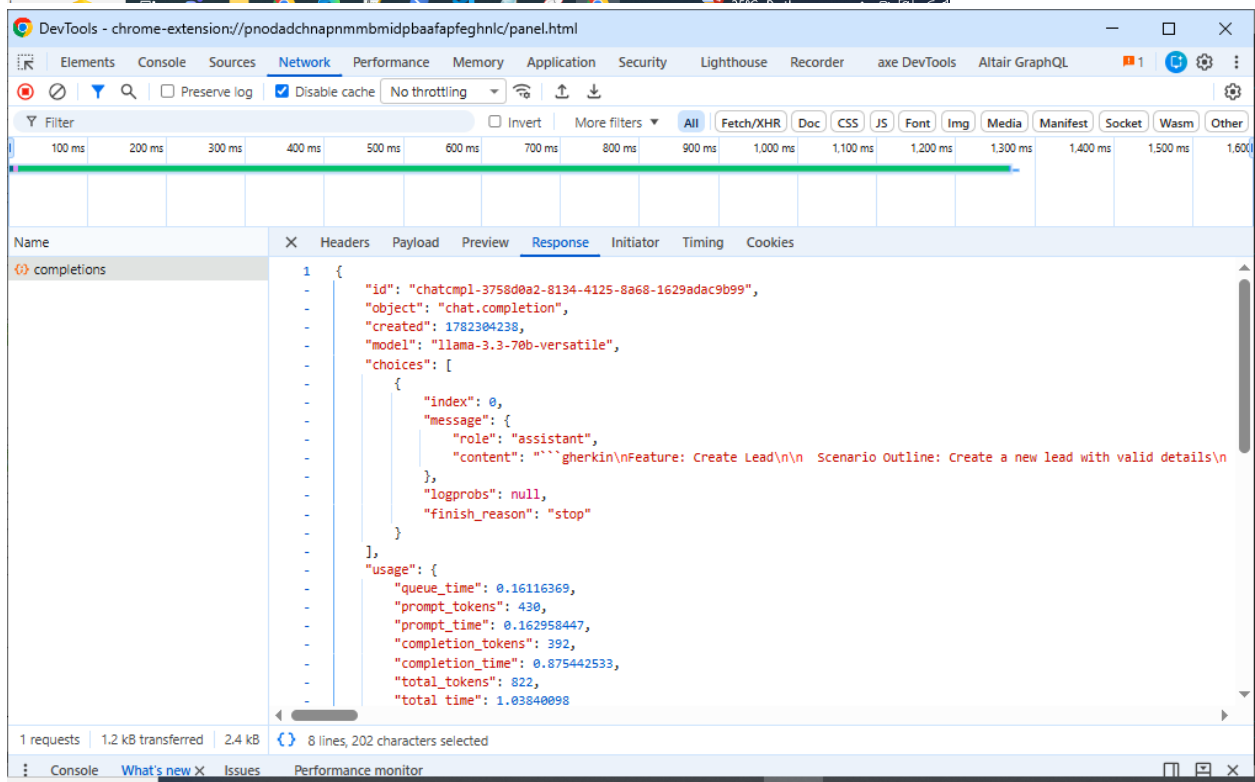
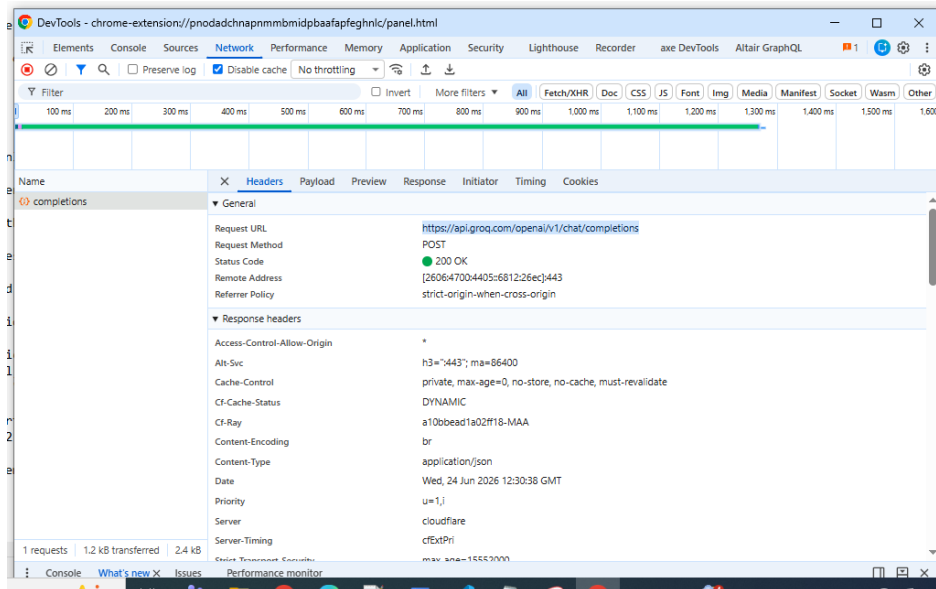
@When("I type {string} into the First Name field")
public void enterFirstName(String firstName) {
    WebElement el =
wait.until(ExpectedConditions.elementToBeClickable(By.id("createLeadForm_firstName")));
    el.sendKeys(firstName);
}

@When("I type {string} into the Last Name field")
public void enterLastName(String lastName) {
    WebElement el =
wait.until(ExpectedConditions.elementToBeClickable(By.id("createLeadForm_lastName")));
    el.sendKeys(lastName);
}

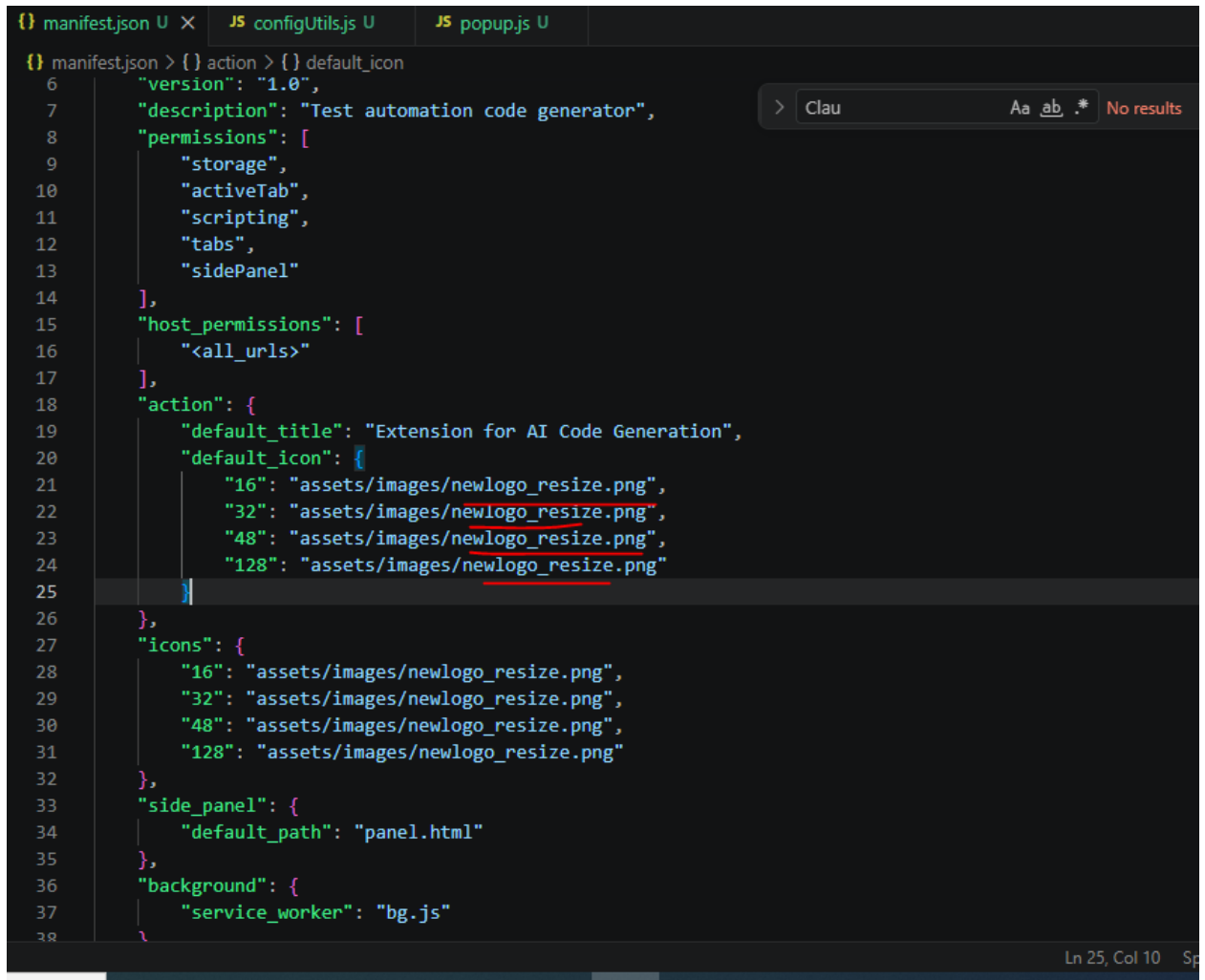
@When("I click the Create Lead button")
public void clickCreateLead() {
    driver.findElement(By.id("ext-gen524")).click();
}

@Then("I should see the created lead details")
public void verifyLeadCreation() {
    WebElement success =
wait.until(ExpectedConditions.visibilityOfElementLocated(By.xpath("//div[@class='x-panel-body x-panel-body-noheader x-panel-body-noborder']")));
    assertTrue(success.isDisplayed());
}

@Then("I should see an error message")
public void verifyErrorMessage() {
    WebElement error =
wait.until(ExpectedConditions.visibilityOfElementLocated(By.xpath("//div[@class='errorMessage']")));
    assertTrue(error.isDisplayed());
}
}
```



Update logo:



The image shows a VS Code editor window with a dark theme. At the top, there are three tabs: 'manifest.json U', 'JS configUtils.js U', and 'JS popup.js U'. The 'manifest.json' tab is active, showing a JSON file. A search bar is located at the top right of the editor, with the text 'Clau' entered, and 'Aa ab .*' and 'No results' are also visible. The JSON content is as follows:

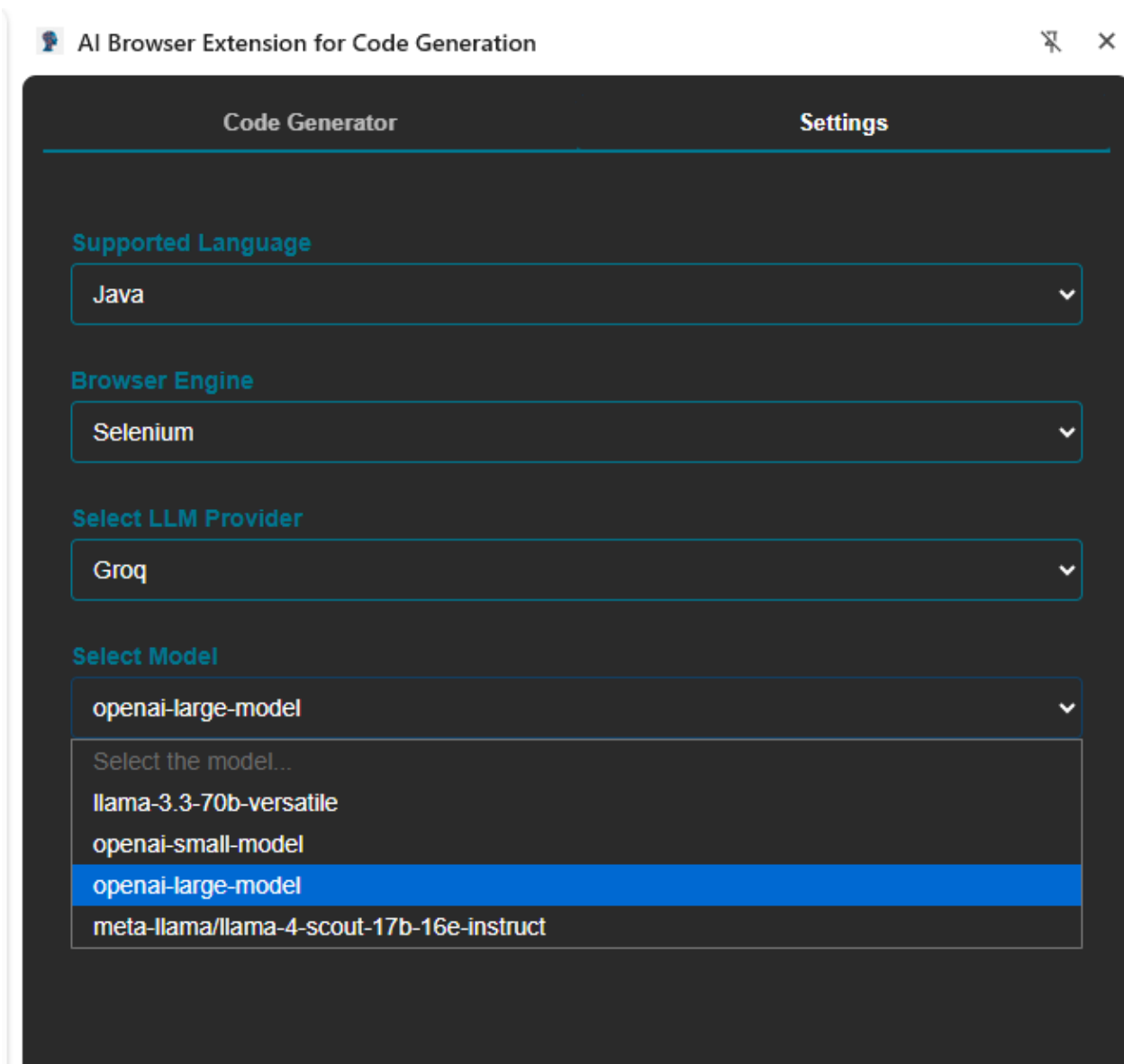
```
{
  "version": "1.0",
  "description": "Test automation code generator",
  "permissions": [
    "storage",
    "activeTab",
    "scripting",
    "tabs",
    "sidePanel"
  ],
  "host_permissions": [
    "<all_urls>"
  ],
  "action": {
    "default_title": "Extension for AI Code Generation",
    "default_icon": {
      "16": "assets/images/newlogo_resize.png",
      "32": "assets/images/newlogo_resize.png",
      "48": "assets/images/newlogo_resize.png",
      "128": "assets/images/newlogo_resize.png"
    }
  },
  "icons": {
    "16": "assets/images/newlogo_resize.png",
    "32": "assets/images/newlogo_resize.png",
    "48": "assets/images/newlogo_resize.png",
    "128": "assets/images/newlogo_resize.png"
  },
  "side_panel": {
    "default_path": "panel.html"
  },
  "background": {
    "service_worker": "bg.js"
  }
}
```

The status bar at the bottom right indicates 'Ln 25, Col 10 Sp'.

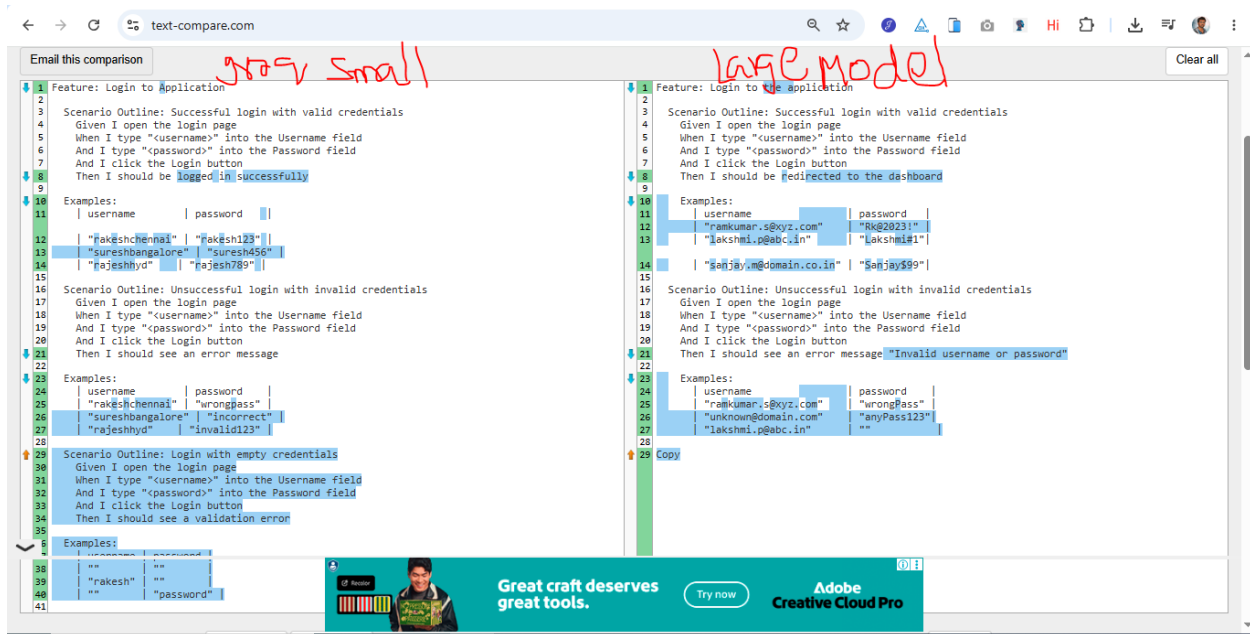
```
src > config > JS appConfig.js > ...
1 export const appConfig = {
2   appName: 'AI Code Generator',
3   logoPath: 'assets/images/newlogoicon.png',
4   colors: {
5     primary: '#ff6b2b',
6     primaryLight: '#234067',
7     accent: 'rgb(43, 116, 142)',
8     success: '#920c2e',
9     successDark: '#a04549',
10    danger: '#ff0000',
11    darkBg: '#2a2a2a',
12    panelBg: '#3a3a3a',
13    mutedText: '#850e1a',
14    // Inspector toggle specific colors
15    gradientStart: '#275fa7',
16    shadowColor: 'rgba(255,107,43,0.3)',
17    shadowColorHover: 'rgba(255,107,43,0.4)',
18    shadowColorActive: 'rgba(255,107,43,0.1)',
19    borderWhite: 'rgba(255,255,255,0.1)'
20  }
21 };
```

Update model in dropdown list

```
src > scripts > JS popup.js > document.addEventListener("DOMContentLoaded") callback
5 document.addEventListener('DOMContentLoaded', async () => {
100 const providerSelect = document.getElementById('providerSelect');
101 const modelSelect = document.getElementById('modelSelect');
102 const groqApiKeyInput = document.getElementById('groqApiKey');
103 const openaiApiKeyInput = document.getElementById('openaiApiKey');
104 const testleafApiKeyInput = document.getElementById('testleafApiKey');
105
106 // Model options by provider
107 const modelsByProvider = {
108   groq: [
109     { value: 'llama-3.3-70b-versatile', label: 'llama-3.3-70b-versatile' },
110     { value: 'openai/gpt-oss-20b', label: 'openai-small-model' },
111     { value: 'openai/gpt-oss-120b', label: 'openai-large-model' },
112     { value: 'meta-llama/llama-4-scout-17b-16e-instruct', label: 'meta-llama/llama-4-scout-17b-16e-instruct' }
113   ],
114   openai: [
115     { value: 'ft:gpt-4o-2024-08-06:personal::D1dF7CWg', label: 'fine tuned' },
116     { value: 'ft:open-mistral-7b:83a13a2e:20251224:43c7eec0', label: 'ft:open-mistral-7b:83a13a2e:20251224:43c7eec0' }
117   ],
118   testleaf: [
119     { value: 'ft:gpt-4o-2024-08-06:qeagle-1::D1QRdvhi', label: 'finetunedmodel' },
120     { value: 'testleaf-gpt-3.5-turbo', label: 'Testleaf GPT-3.5 Turbo' },
121     { value: 'ft:gpt-4o-mini-2024-07-18:qeagle-1::DDsfzagZ', label: 'ft:gpt-4o-mini-2024-07-18:qeagle-1::DDsfzagZ' }
122   ],
123   { value: 'ft:gpt-4o-mini-2024-07-18:qeagle-1::DEDfKW4y', label: 'Finetuned model' },
124
125   //ft:gpt-4o-mini-2024-07-18:qeagle-1::D7bXRJPX
126
127 ]
128 };
129
130 };
```



Comparison Feature file and page class with small vs large llama model

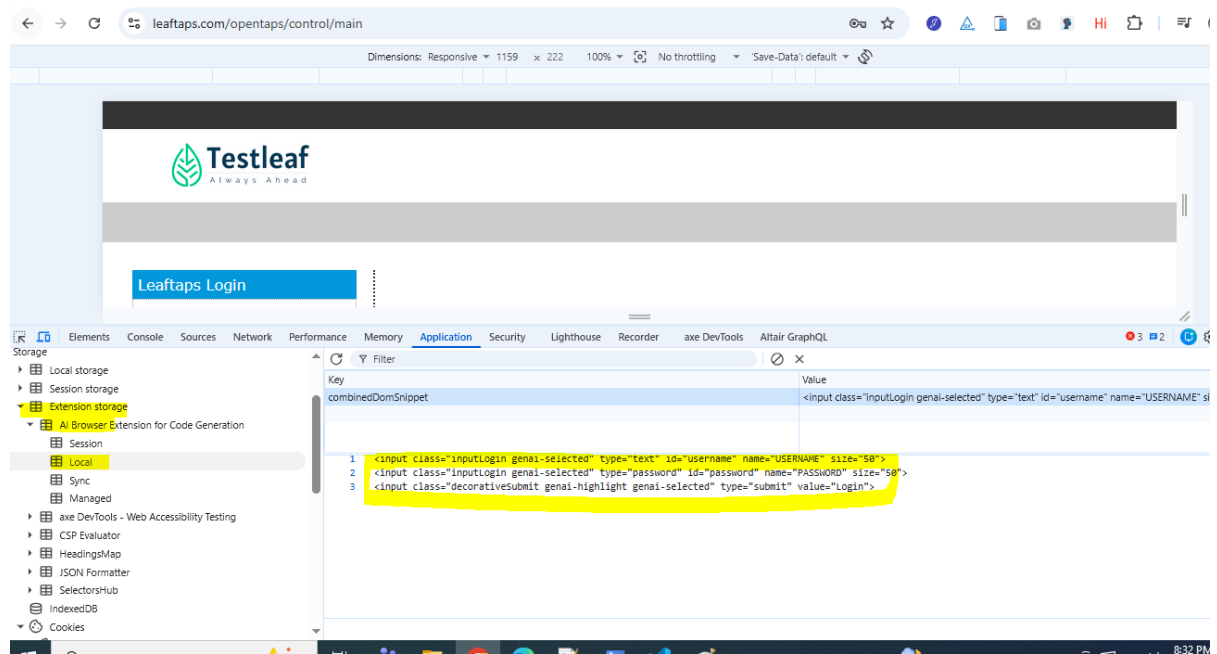


Root file, assest file, lib files:

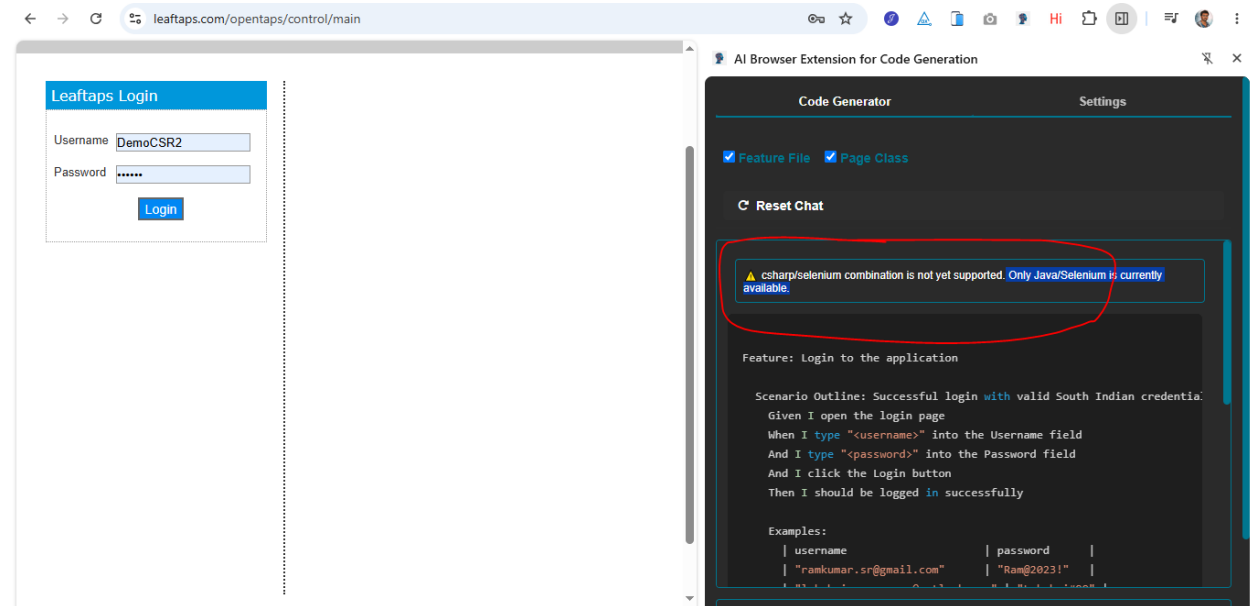
1. manifest.json - extension identify & permission.
2. bg.js - background workers always running
3. Panel.html - The side panel UI
4. Assest/image - tools icon or logo image
5. Lib/marked/**marked.min.js** - **Markdown → HTML**
6. prism/
 - prism.js - syntax highlighter core
 - prism-typescript.js - typescript highlighter
 - prism-gherkin-min.js - gherkin highlighter
 - prism-tomorrow.css - Dark code theme
- src
7. Src/config
 - appconfig.js - Brand color
 - configutils.js - Applies colors to CSS
8. Content/content.js - Injected into webpage
9. Scripts/
 - init-color.js - Prevents white flash
 - init.config.js - Initializes config
 - tabs.js - Tab switching
 - popup.js - Settings tab
 - chat.js - Core engine
 - prompt.js - AI prompt templates
10. Api/groq-api.js - groq connector

Inspected DOM html in local storage

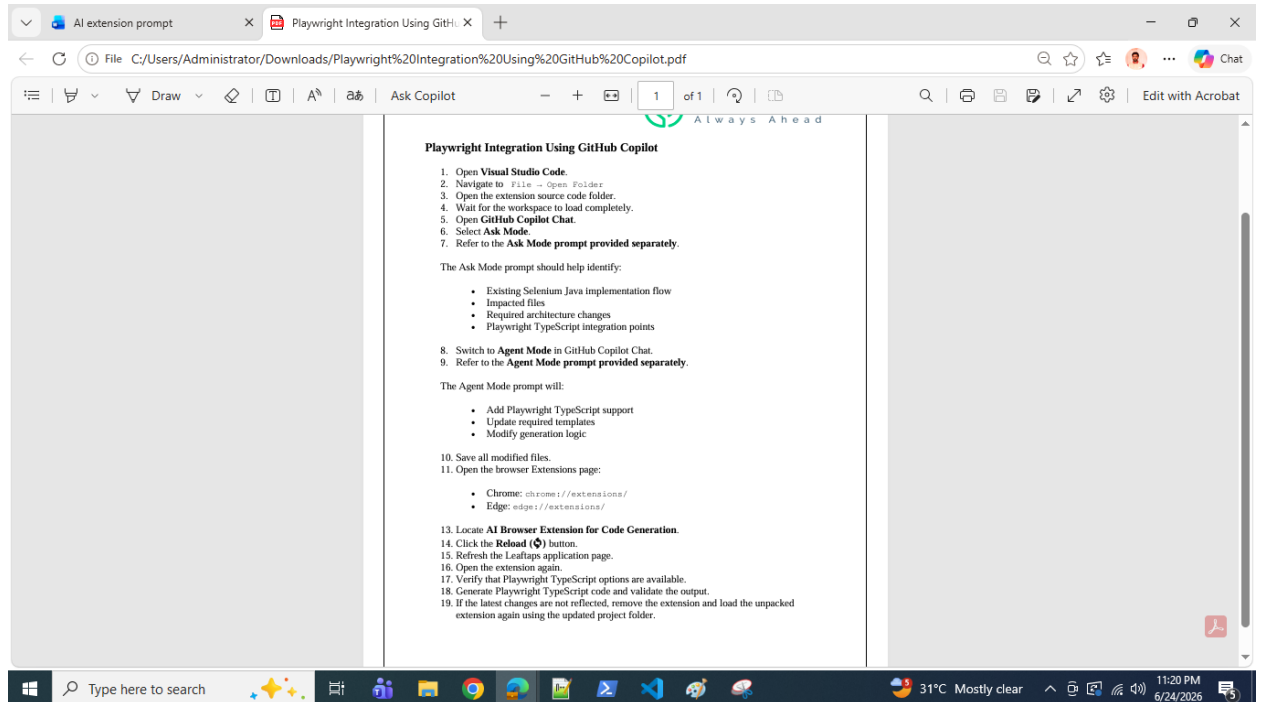
Inspect -> applications -> Extension name -> local storage



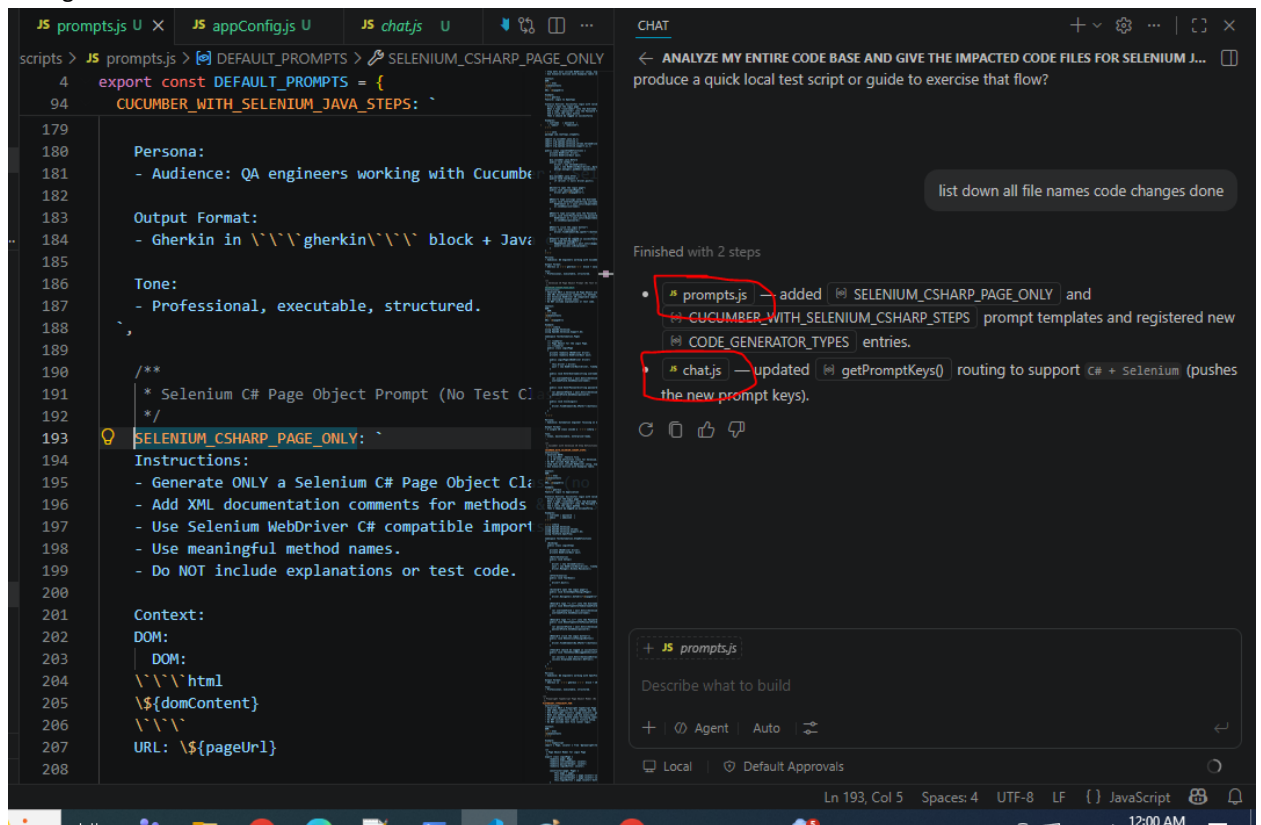
Selenium C#



Follow below steps



Changes done in below files



C:\Users\Administrator\Downloads\ai-extension\ai-extension\src\scripts\chat.js

Ln: 1 Col: 1/51 Ch: 1/51

UTF-8

Unix

```
// Generate appropriate prompt keys based on selections and language
if (isFeatureChecked && isPageChecked) {
  // Both feature and page selected - generate combined output
  if (this.isJavaSelenium(lang, eng)) {
    promptKeys.push('CUCUMBER_WITH_SELENIUM_JAVA_STEPS');
  } else if (this.isTypeScriptPlaywright(lang, eng)) {
    promptKeys.push('PLAYWRIGHT_TYPESCRIPT_POM_AND_SPEC');
  } else {
    // For non-Java/Selenium and non-TypeScript/Playwright combinations
    promptKeys.push('CUCUMBER_ONLY');
    this.addUnsupportedLanguageMessage(lang, eng);
  }
} else if (isFeatureChecked) {
  // Feature file only
  if (this.isTypeScriptPlaywright(lang, eng)) {
    promptKeys.push('PLAYWRIGHT_TYPESCRIPT_SPEC');
  } else {
    // Feature file only
    if (this.isJavaSelenium(lang, eng)) {
      promptKeys.push('SELENIUM_JAVA_PAGE_ONLY');
    } else if (this.isTypeScriptPlaywright(lang, eng)) {
      promptKeys.push('PLAYWRIGHT_TYPESCRIPT_POM');
    } else {
      this.addUnsupportedLanguageMessage(lang, eng);
    }
  }
}
```

D:\AI_Program\AI_TL\CourselImages\Week3\ai-extension\ai-extension\src\scripts\chat.js

Ln: 1 Col: 1/51 Ch: 1/51

UTF-8

Unix

```
// Generate appropriate prompt keys based on selections and language/e
if (isFeatureChecked && isPageChecked) {
  // Both feature and page selected - generate combined output
  if (this.isJavaSelenium(lang, eng)) {
    promptKeys.push('CUCUMBER_WITH_SELENIUM_JAVA_STEPS');
  } else if (this.isCSharpSelenium(lang, eng)) {
    promptKeys.push('CUCUMBER_WITH_SELENIUM_CSHARP_STEPS');
  } else if (this.isTypeScriptPlaywright(lang, eng)) {
    promptKeys.push('PLAYWRIGHT_TYPESCRIPT_POM_AND_SPEC');
  } else {
    // For other combinations, fall back to Cucumber-only and warn
    promptKeys.push('CUCUMBER_ONLY');
    this.addUnsupportedLanguageMessage(lang, eng);
  }
} else if (isFeatureChecked) {
  // Feature file only
  if (this.isTypeScriptPlaywright(lang, eng)) {
    promptKeys.push('PLAYWRIGHT_TYPESCRIPT_SPEC');
  } else {
    // Feature file only
    if (this.isJavaSelenium(lang, eng)) {
      promptKeys.push('SELENIUM_JAVA_PAGE_ONLY');
    } else if (this.isCSharpSelenium(lang, eng)) {
      promptKeys.push('SELENIUM_CSHARP_PAGE_ONLY');
    } else if (this.isTypeScriptPlaywright(lang, eng)) {
      promptKeys.push('PLAYWRIGHT_TYPESCRIPT_POM');
    } else {
      this.addUnsupportedLanguageMessage(lang, eng);
    }
  }
}
```

C:\Users\Administrator\Downloads\ai-extension\ai-extension\src\scripts\prompts.js

Ln: 1 Col: 1/4 Ch: 1/4

Windows-1252

Unix

```
- Professional, executable, structured.
```

D:\AI_Program\AI_TL\CourselImages\Week3\ai-extension\ai-extension\src\scripts\prompts.js

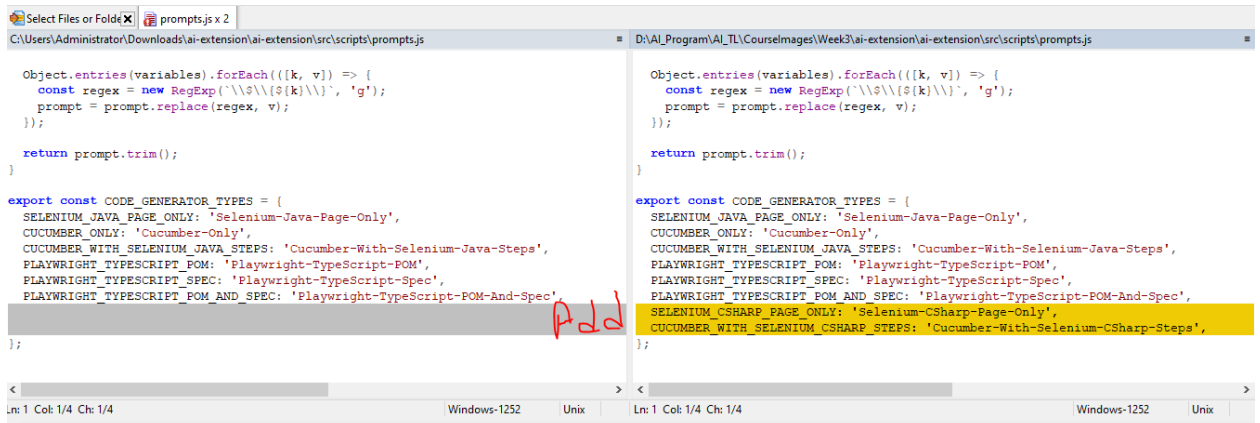
Ln: 1 Col: 1/4 Ch: 1/4

Windows-1252

Unix

```
/**
 * Selenium C# Page Object Prompt (No Test Class)
 */
SELENIUM_CSHARP_PAGE_ONLY: `
Instructions:
- Generate ONLY a Selenium C# Page Object Class (no test code).
- Add XML documentation comments for methods & class.
- Use Selenium WebDriver C# compatible imports.
- Use meaningful method names.
- Do NOT include explanations or test code.

Context:
DOM:
  DOM:
  \\\`html
  \${domContent}
  \\\`
URL: \${pageUrl}`
```



Feature: LeafTaps login functionality

As a QA engineer

I want to verify that users can log in with valid credentials
So that they are taken to the CRM/SFA home page

Scenario Outline: Successful login with valid credentials

Given I navigate to the LeafTaps login page

When I enter "<username>" into the Username field

And I enter "<password>" into the Password field

And I click the Login button

Then I should see the CRM/SFA home page

Examples:

username	password
DemoSalesManager	crmsfa
admin	admin123

using System;
using OpenQA.Selenium;
using OpenQA.Selenium.Chrome;
using OpenQA.Selenium.Support.UI;
using SeleniumExtras.WaitHelpers;
using TechTalk.SpecFlow;
using FluentAssertions;

namespace TestAutomation.StepDefinitions

```
{
  [Binding]
  public class LoginSteps
  {
    private IWebDriver _driver;
    private WebDriverWait _wait;
    private const string LoginUrl = "https://leaf taps.com/opentaps/control/main";
```

#region Hooks

[BeforeScenario]

```

public void Setup()
{
    var chromeOptions = new ChromeOptions();
    // Uncomment the line below to run head-less (CI environments)
    // chromeOptions.AddArgument("--headless");
    _driver = new ChromeDriver(chromeOptions);
    _driver.Manage().Window.Maximize();

    // Explicit wait – 10 seconds timeout, polling every 500 ms
    _wait = new WebDriverWait(_driver, TimeSpan.FromSeconds(10))
    {
        PollingInterval = TimeSpan.FromMilliseconds(500)
    };
    _wait.IgnoreExceptionTypes(typeof(NoSuchElementException),
typeof(StaleElementReferenceException));
}

[AfterScenario]
public void TearDown()
{
    _driver?.Quit();
}

#endregion

#region Step Definitions

[Given(@"I navigate to the LeafTaps login page")]
public void GivenINavigateToTheLeafTapsLoginPage()
{
    _driver.Navigate().GoToUrl(LoginUrl);
}

[When(@"I enter ""(.*)"" into the Username field")]
public void WhenIEnterIntoTheUsernameField(string username)
{
    var usernameField = _wait.Until(
        ExpectedConditions.ElementIsVisible(By.Id("username"))
    );
    usernameField.Clear();
    usernameField.SendKeys(username);
}

[When(@"I enter ""(.*)"" into the Password field")]
public void WhenIEnterIntoThePasswordField(string password)
{
    var passwordField = _wait.Until(
        ExpectedConditions.ElementIsVisible(By.Id("password"))
    );
    passwordField.Clear();
    passwordField.SendKeys(password);
}

```

```

    }

    [When(@"I click the Login button")]
    public void WhenIClickTheLoginButton()
    {
        // The login button is an <input type="submit" value="Login">
        var loginButton = _wait.Until(
            ExpectedConditions.ElementToBeClickable(By.CssSelector("input.decorativeSubmit"))
        );
        loginButton.Click();
    }

    [Then(@"I should see the CRM/SFA home page")]
    public void ThenIShouldSeeTheCrmSfaHomePage()
    {
        // After a successful login the page shows a link with text "CRM/SFA"
        var crmLink = _wait.Until(
            ExpectedConditions.ElementIsVisible(By.LinkText("CRM/SFA"))
        );
        crmLink.Displayed.Should().BeTrue("the CRM/SFA link must be visible after a successful login");
    }

    #endregion
}
}

```